POSTER PRESENTATION ON WEB PROGRAMMING AND MOBILE APPLICATION DEVELOPMENT PROJECT

FleetSync

A Multi-Tenant SaaS Platform for Real-Time Vehicle Tracking and Fleet Management

Sidratul Muntaha (210208) · G M Amanullah (220225)
drik2108@cseku.ac.bd · gmamanullah21@gmail.com

Supervisor: Dr. Kazi Masudul Alam, Professor
Computer Science and Engineering Discipline
Khulna University

Abstract

Institutions still track vehicles with phone calls and spreadsheets, while commercial telematics needs dedicated hardware. FleetSync is a self-service, multi-tenant SaaS platform that tracks vehicles live using only the driver's smartphone. A Node.js API with JWT roles keeps each tenant's data isolated, WebSockets push updates instantly, and PostgreSQL/PostGIS stores trips and GPS points. A GPS fix reached the passenger in a median of 9 ms, and all 73 API tests passed.

Introduction

Campus buses, delivery vans and office shuttles move people every day, yet riders rarely know where their vehicle is. Samsara and Geotab serve large enterprises with in-vehicle hardware. FleetSync targets everyone else: the driver's phone becomes the tracker, and each institution gets its own isolated workspace.

Problem Statement

- Riders can't see where the vehicle is.
- Route and driver changes arrive late.
- Trackers are costly; always-on phones hurt driver privacy.
- Shared systems must keep each tenant's data apart.

Objectives

Multi-tenant — Isolated data per institution

Three roles — Authority, driver, passenger

No hardware — GPS from the driver's phone

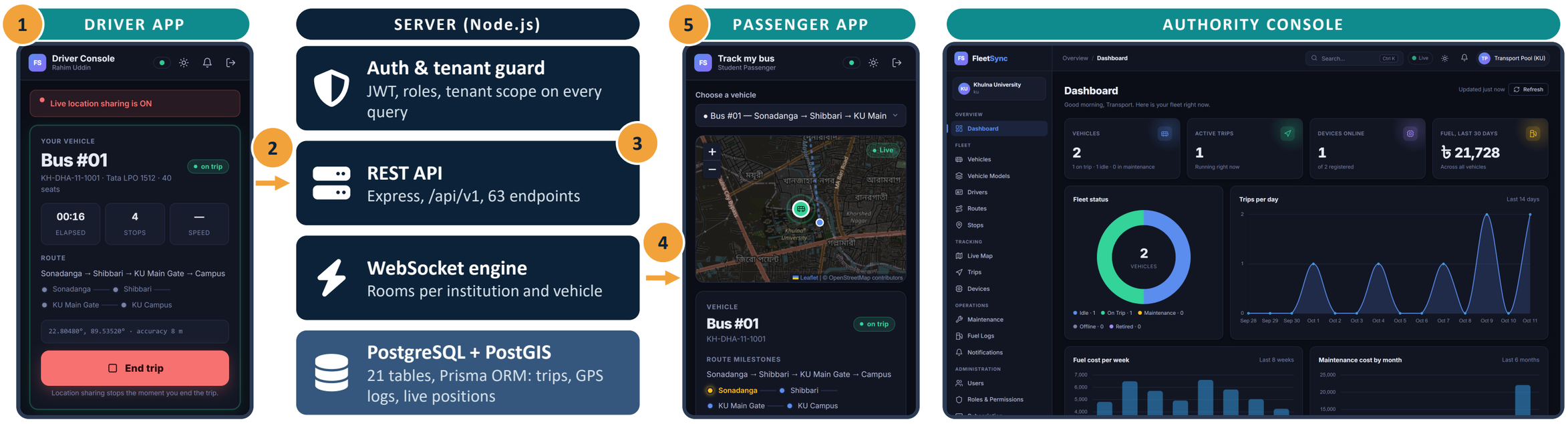
Real-time — Instant sync over WebSockets

Private — Tracking only during a trip

Operations — Maintenance, fuel and alerts

Methodology

Node.js + Express · JWT + bcrypt · PostgreSQL + PostGIS · WebSockets (ws) · Leaflet + OSM



1. Driver taps Start trip.
2. Phone sends a GPS fix (HTML5 Geolocation).
3. API checks role, tenant and active trip, then saves the point.
4. WebSocket engine pushes location.update.
5. Passenger and authority maps move. End trip stops sharing.

Dataset & test setup. Seeded multi-tenant data: 3 institutions, 10 users, 4 drivers, 4 vehicles, 4 routes (12 stops), trips with GPS history, fuel and maintenance logs. API and PostgreSQL 18 / PostGIS 3.6 ran on one laptop (Core i7-10510U, 8 GB RAM), so timings exclude mobile-network delay.

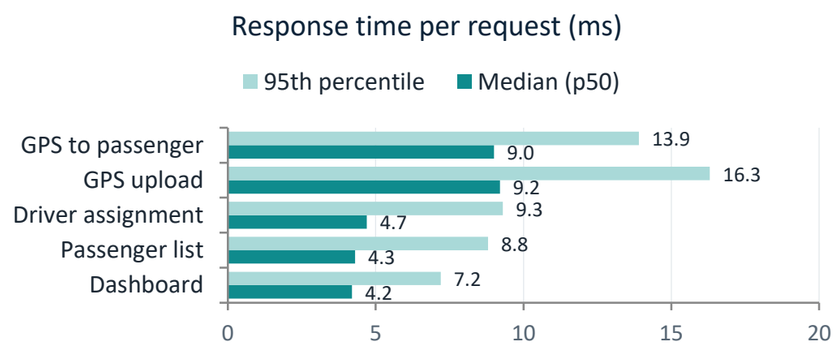
Experimental Results

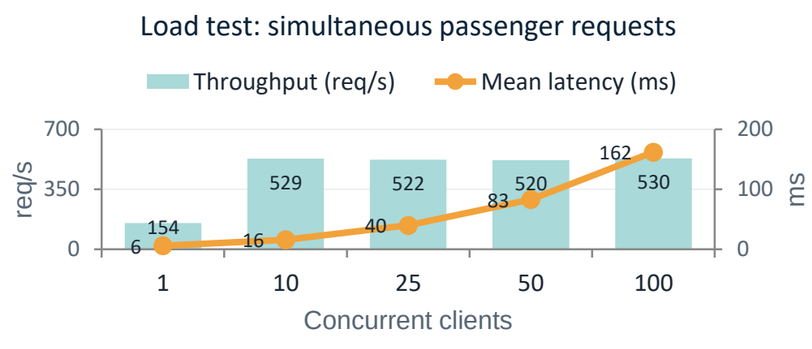
9 ms — median GPS delay, driver to passenger

40 / 40 — GPS updates delivered live

73 / 73 — API tests passed (17 on tenant isolation)

≈525/s — requests/s, 700/700 OK at 100 clients





Privacy check: 20 of 20 GPS uploads from an off-duty driver were rejected. Throughput levels off near 525 req/s, then latency grows linearly. Login takes 77 ms by design (bcrypt).

Conclusion and Future Work

FleetSync shows that a phone-based, multi-tenant platform can deliver live tracking and instant updates without dedicated hardware. Every query is tenant-scoped, and tracking is tied to an active trip.

Next steps

- Android app with background GPS
- ETA prediction from trip history
- Offline GPS buffering in weak signal
- Cloud deployment and field testing

Source Code

Scan the QR code or visit

github.com/smdrik007/FleetSync